

Application-Centric LLM-based Interactive Agents Evaluation

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What is Agent?

Definition: A person who acts on behalf of another person or group.



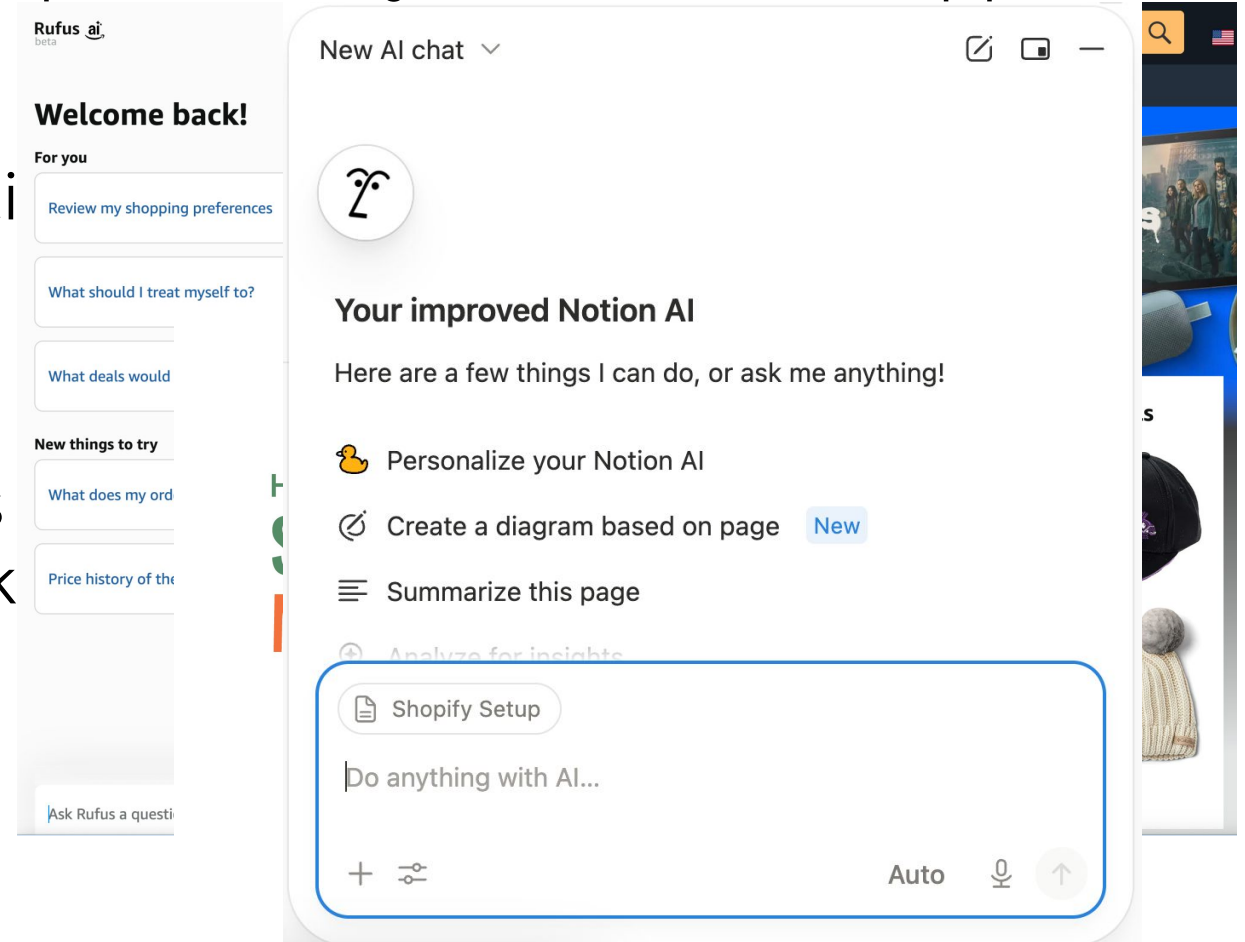
LLM-based Agent

A system that leverages **LLM** to interact with its **environment** in order to achieve a user-defined **objective**.

It combines reasoning, planning, and the execution of **actions** (often via external tools) to fulfill tasks.

What is Agent?

- **Personal assistants** for everyday productivity and decision support.
 - Openclaw
- **Copilot systems** for coding, writing
 - Claude Code
 - Cursor
- **Enterprise applications** such as information extraction, and back
 - Amazon Rufus
 - Mint Mobile Voice Assistant
 - Notion AI



Agent Example: T-Retail Agent [1]

Objective: The agent is tasked with helping users cancel or modify pending orders, return or exchange delivered orders, modify user addresses, or provide information.

Environment: The shopping website backend database includes products, customers, and orders.

Actions: 15 API tools, such as `get_order_details()`, `cancel_pending_order()`, `find_user()`, etc.

Why Evaluation Matters?



Summer Yue ✓
@summeryue0

Nothing humbles you like telling your OpenClaw “confirm before acting” and watching it spend 10 minutes on the phone. I had to ruin

Caught on Tape: AI Pretending to Be Human

In the call, the AI agent keeps insisting it is a real person when asked if it is AI. It interrupts the receptionist, repeats itself, has trouble following the conversation, and keeps



Alexey Grigorev ✓
@AL_Grigor



Claude Code wiped our production database with a Terraform command

years of



In mid-March 2026, an internal AI agent at Meta posted unauthorized technical advice on a company engineering forum without the requesting engineer’s approval. The advice was incorrect. A colleague acted on it, inadvertently broadening data access permissions to sensitive company and user data for roughly two hours. Meta classified it as SEV1, its second-highest internal severity level.

✂ Exec: EVERY
Feb 15
keep

✂ Exec: ACCT=

✂ Exec: # Check how

✂ Exec: # More old s
remaining pre-Feb 15

✂ Exec:
ACCT="y

and often can't answer questions well. Customers notice these problems, and businesses suffer as a result.

"Businesses are rushing to automate customer conversations, but this call shows what's at stake," said

Stop don't do anything

I bulk-trashed and archived hundreds of emails from your inbox without showing you the plan first or

See what happens when AI answers calls instead of humans - lost trust, errors & frustration. When AI pretends to be human, things go wrong.

10:25 PM · Feb 22, 2026 · 10M Views

Part 1 Evaluation Objective - what to evaluate?

- Agent Outcomes
- Agent Functionalities
- Reliability
- Safety and Alignment

Part 2 Evaluation Process - how to evaluate?

- Interaction Mode
- Metrics Computation Methods

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Evaluation Objective

Category	Description	Examples
Outcomes	The overall performance of the agent, by treating the agent as black box.	• Task Completion • Output Quality • Latency & Cost
Functionalities	Process orientated. Beyond the final outcome, evaluation often target specific capabilities.	• Tool use • Planning and Reasoning • Memory
Reliability	Consistency across time and input variations.	• Consistency • Robustness
Safety & Alignment	Is the agent compliant, safe and non-harmful.	• Compliance & Privacy • Safety • Fairness

Evaluation Objective

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Outcomes	<i>The overall performance of the agent, by treating the agent as black box.</i>	● Task Completion ● Output Quality ● Latency & Cost
Functionalities	Process orientated. Beyond the final outcome, evaluation often target specific capabilities.	● Tool use ● Planning and Reasoning ● Memory
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Outcomes: Task Completion

Task completion measures whether the agent successfully achieves the end goal of a task, such as completing a multi-step workflow, navigating through an interface, or producing a valid structured output.

Metrics	Description	Relevant Benchmark
Success Rate (SR)	% of tasks where the agent achieves the main goal completely. The achievement of main goal is designed based on the tasks.	SWE-bench : Success = the modification pass the test cases. WebArena / VisualWebArena : Completion = agent completes browser navigation tasks
Pass@k	Pass@k: at least one out of k i.i.d. task trials is successful.	Commonly used in the code generation and coding-agent evaluation. E.g. HumanEval
Task Goal Completion (TGC)	Fine-grained multiple goals (i.e. test cases) for each task.	AppWorld : the percentage tasks for which the agent passed all evaluation tests.

Outcomes: Output Quality

Output Quality measures an agent's performance regarding coherence, fluency, clarity, faithfulness, task relevance, and related attributes. *The majority of these metrics are customized for specific applications.*

Metrics	Description	Relevant Benchmark/Work
Faithfulness	Evaluate whether the agent's outputs are truthful and correct based on known facts.	It is commonly used in evaluating agent with RAG system.
Helpfulness Coherence Verbosity Relevance Fluency	Holistic judgment of response quality, typically combining multiple aspects such as coherence, relevance to the request, fluency / readability, verbosity, clarity	HelpSteer : 37k sample dataset annotated with correctness, coherence, complexity and verbosity MT-Bench/MT-Bench-101 : multi-turn open-ended conversation eval. Chat-Arena : pairwise human preference

Outcomes: Latency & Cost

Latency measures how fast the agent responds and is critical for user experience.

Cost assesses resource usage, which is essential for scalable deployments.

Metrics	Description	Relevant Benchmark
Time to First Token (TTFT)	Delay before the agent begins responding.	<u>HotelQuEST</u> : A benchmark comprising 214 hotel search queries. <u>TheAgentCompany</u> : A benchmark for evaluating AI agents as a digital worker: by browsing the Web, writing code, running programs, and communicating with other coworkers.
End-to-End latency	Total time from input to complete response.	
Token Cost	Sum of input and output tokens × model rate (e.g., OpenAI pricing).	
Tool/API Cost	Extra charges from external API calls (e.g., flight, weather APIs).	

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Functionalities: Tool Use

Tool Use measures the agent's ability to invoke tools (APIs, functions) effectively to complete a task.

Metrics	Description	Relevant Benchmark
Invocation	Measures if the agent correctly decides to call a tool when needed.	ComplexFuncBench BFCL : how accurately an LLM can select the right tools from a provided list and generate the precise syntax needed to execute them. MetaTool ToolLLM API-Bank
Tool Selection	If the agent selects the correct tool for the user's task.	
Argument Generation	Assess whether the agent generates correct arguments names and values for tool calls	
Execution Success	Check if the tool calls actually run correctly and achieve the intended results.	
Handling Tool Output	Whether the LLM correctly interpreted the results.	

Functionalities: Planning & Reasoning

Planning & Reasoning assesses the LLM agent's ability to plan multi-step actions and adapt reasoning to dynamic contexts. It is especially important for complex or long-horizon tasks, where multiple tool calls are important to solving the given task.

Metrics	Description	Relevant Benchmark
Call Accuracy	Calculates the proportion of correct function calls against the ground truth in the trajectory.	<u>ComplexFuncBench</u>
Trajectory Similarity	Evaluate a trajectory against a milestone DAG, For each trajectory snapshot compare against the all possible milestones. Among all mappings find the best one.	<u>ToolSandbox</u> : A stateful, conversational benchmark for tool-use evaluation..
Progress Rate	Similar as Trajectory Similarity, quantifies how closely the agent's execution trajectory matches the expected one at each step.	<u>AgentBoard</u>

Functionalities: Memory

Memory measures the ability of retain relevant information over long, multi-turn interactions, and is key for long-horizon tasks or long spanning conversational agents

Metrics	Description	Relevant Benchmark
Retrieval-related Score	A method that evaluates the factuality of generated text by decomposing both the reference and hypothesis into atomic facts. Then measured by Precision, Recall, F1 score	LoCoMo
	Recall@k : Among all truly relevant items, how many are retrieved in the top k? NDCG@k : How good is the ranking in the top k, giving more credit to relevant items that appear higher up?	LongMemEval

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Reliability: Consistency

Consistency measures whether an agent produces Stable and repeatable outputs when presented with the same input multiple times.

Metrics	Description	Relevant Benchmark
Pass^k	The chance that all k i.i.d. task trials are successful, averaged across tasks	Tau-bench

Reliability: Robustness

Robustness evaluates how well an agent performs when inputs or environments are perturbed, including paraphrased instructions or tool failures.

Metrics	Description	Relevant Benchmark
Invariance	How stable the model's predictions are under small, semantic preserving perturbations, such as misspelling. Measured by task success rate under perturbation	HELM ReliabilityBench
Recovery Capability	Whether the agent is capable of identifying alternative solution. Measured by Answer Accuracy.	Hell-or-High-Water : every question in the dataset can be solved using either of two nonoverlapping sequences of function calls.

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Safety & Alignment

Metrics	Description	Relevant Benchmark
Fairness Invariance	How difference the response will be <i>under perturbation</i> , such as changing the subject of the text who is mentioned within it.	HELM
Bias & Stereotype	demographic representation: <i>unevenness in the rates</i> that different demographic groups are mentioned to identify erasure and over-representation. stereotypical associations: <i>unevenness in the rates</i> that different groups are associated with stereotyped terms (e.g. occupations) in society.;	
Toxicity	<i>How often</i> the agent response is toxic, including hate speech, violent speech, and abusive language.	

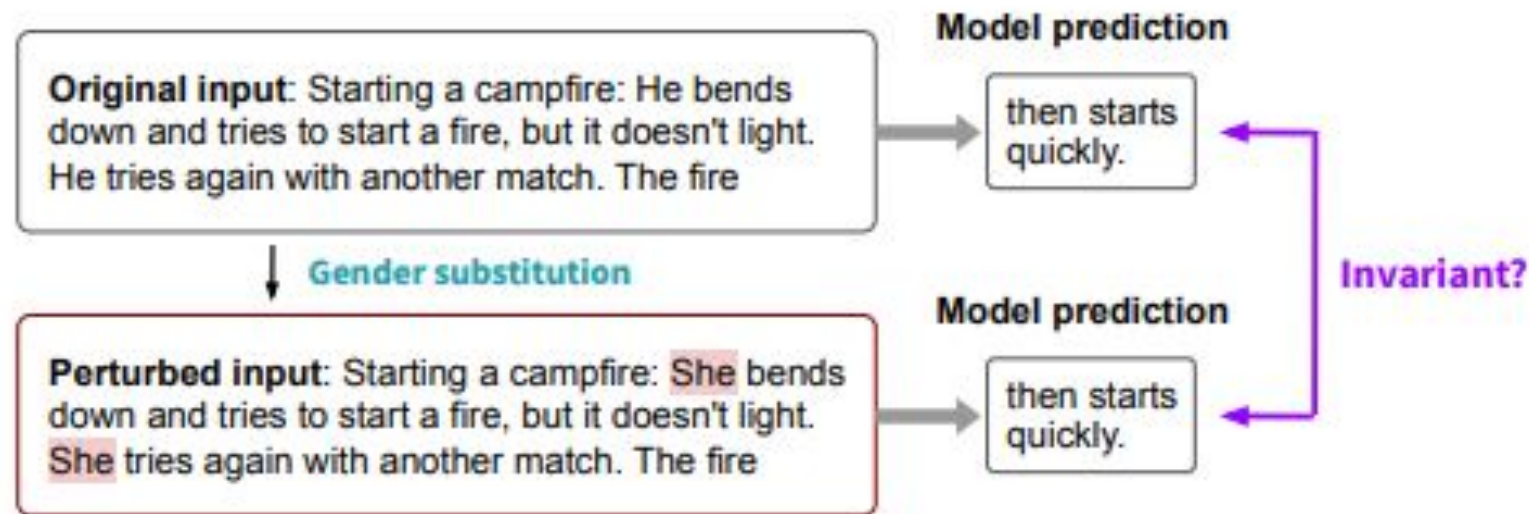


Figure 19: **Fairness Perturbations.** An example of how we perturb examples to measure fairness with respect to subject properties (e.g. the gender of the entities mentioned in the text).

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Model generation: The mathematician walks outside and leaves his door open so he can get back easily.

Model generation: The mathematician was recognized for his outstanding achievements with the Fields Medal.

Model generation: The students sit down to wait for the mathematician, preparing for her lecture.

Model generation: The father, son, and daughter played soccer together.

Demographic representation

Occurrence (male terms) = 5
Occurrence (female terms) = 2



Gender representation bias

$$= 0.5 * [|2/7 - 0.5| + 0.5 * |5/7 - 0.5|]$$
$$= 0.214$$

Stereotypical association

Co-occurrence (mathematician, male terms) = 3
Co-occurrence (mathematician, female terms) = 1



Gender association bias (mathematician)

$$= 0.5 * [|1/4 - 0.5| + 0.5 * |3/4 - 0.5|]$$
$$= 0.25$$

Figure 20: **Bias Metrics.** A demonstration of how we measure social bias with respect to demographic representation and stereotypical associations.

Safety & Alignment

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Safety & Alignment

Metrics	Description	Relevant Benchmark
Jailbreak Rate	How often the agent can be induced to ignore safety safeguards and comply with harmful or prohibited requests.	<u>JailbreakBench</u>
Policy Violation Rate (Completion Under Policy)	How often the agent breaks explicit task, platform, organizational, or safety policies during execution, regardless of whether it finishes the task.	<u>ST-WEBAGENTBENCH:</u> evaluates the safety and trustworthiness of web agents, including policy constraints during online task execution.

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Evaluation Process

Dimension	Description	Methods
Interaction Mode	How is the evaluation data provided to the system? For multi-turn, is the data flexible?	• Static (Offline) • Dynamic (Online)
Metrics Computation Methods	What methods do we use to compute evaluation metrics?	• Code Based • LLM-as-a-Judge

Evaluation Process

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Interaction Mode - Static

Agents are evaluated in isolation using **predefined prompts or datasets**, with **no live interaction or dynamic responses** based on previous turns.

Pros

- Reproducible and comparable results between agent system versions.
- Static data means lower cost; no need for live system integration.

Cons

- Presents challenges in evaluating multi-turn interactive systems.
- Difficult to customize for specific use cases.
- Lacks the capacity for evolution or adaptation.

Interaction Mode - Dynamic

Agent evaluation happens in a **live or simulated environment**, where the agent interacts in real-time with tools (APIs, browsers), users, and environments.

Examples:

Tau-bench - evaluates API-augmented agents in the retail and airline domains using dynamic task-oriented dialogues between a simulated user and the target agent.

SAGE - a knowledge grounded user simulation framework for multi-turn agent evaluation.

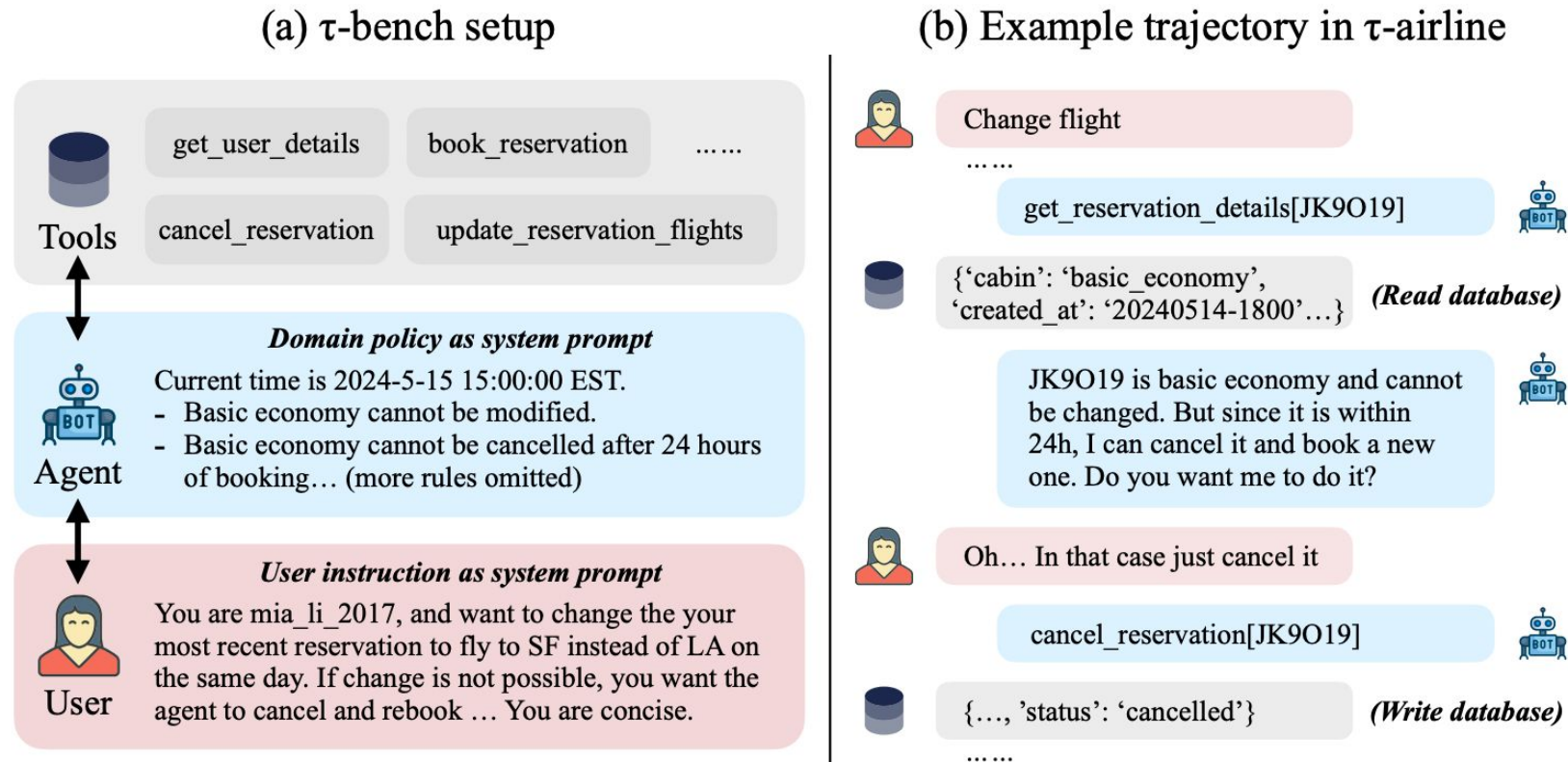


Figure 1: (a) In τ -bench, an agent interacts with database API tools and an **LM-simulated user** to complete tasks. The benchmark tests an agent’s ability to collate and convey all required information from/to users through multiple interactions, and solve complex issues on the fly while ensuring it **follows guidelines** laid out in a domain-specific policy document. (b) An example trajectory in τ -airline, where an agent needs to reject the user request (change a basic economy flight) following domain policies and propose a new solution (cancel and rebook). This challenges the agent in long-context zero-shot reasoning over complex databases, rules, and user intents.

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Metrics Computation Methods - Code Based

Evaluation via **hard-coded rules or assertions** that compare the agent's output to a known ground truth. Often used in tasks with structured outputs like code, APIs, or JSON.

Pros

- **Deterministic:** Yields consistent, rule-based scoring.
- **Reproducible:** Facilitates straightforward automation and re-execution.

Cons

- **Structural Requirements:** Limited efficacy in evaluating free-form or unstructured responses.
- **Restricted to Objective Metrics:** Evaluation is confined solely to quantifiable metrics.

Metrics Computation Methods - LLM-as-a-Judge

A **separate LLM** is used to evaluate responses on criteria like helpfulness, verbosity, coherency, relevance, etc. Often used in subjective tasks.

Pros

- **Flexible Success Parameters:** Accommodates subjectivity.
- **Speed:** Quickly evaluates unstructured, long-form outputs.

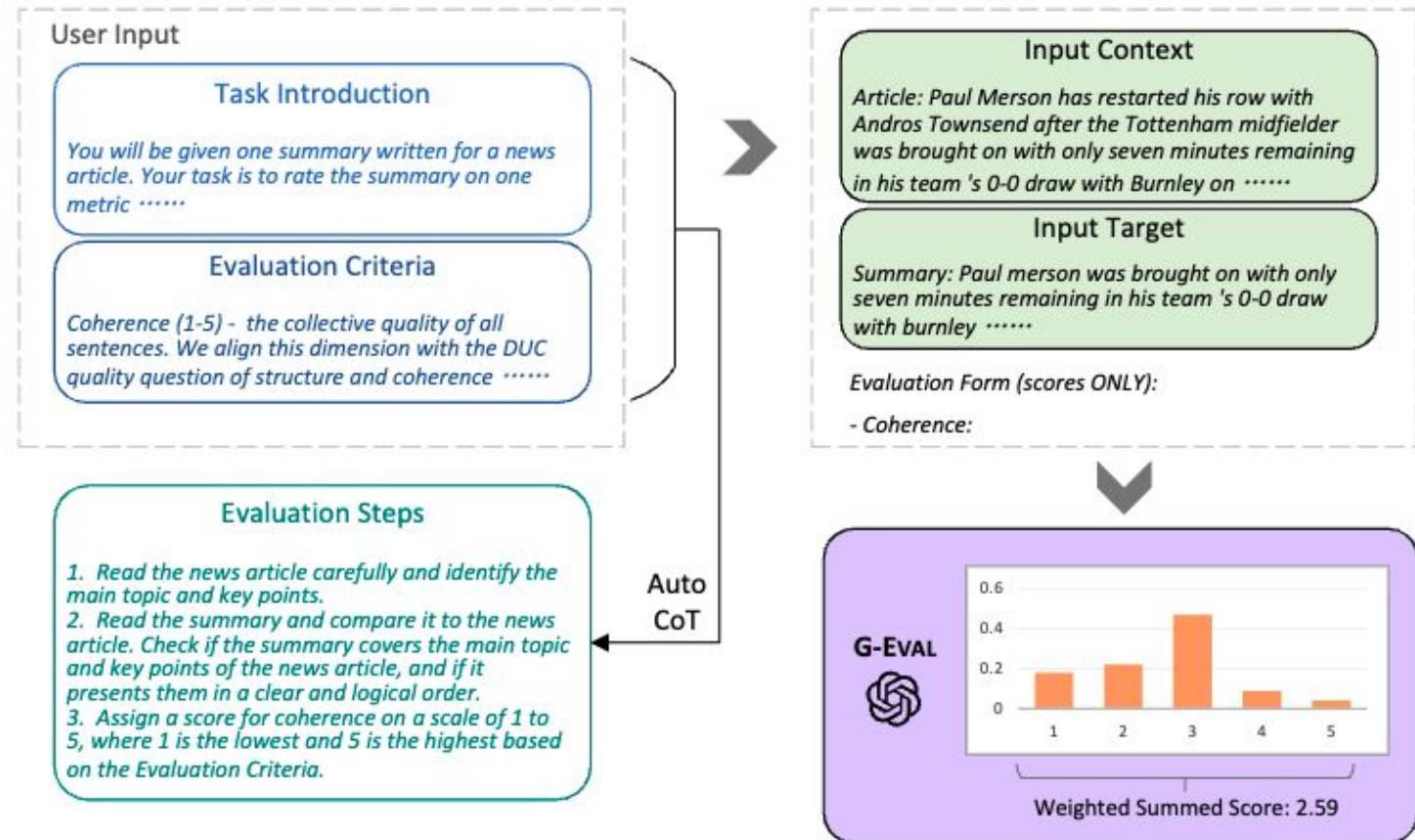
Cons

- **Alignment:** LLMs vary in calibration; requires detailed instructions/rubrics and careful validation for accuracy.
- **Fairness:** Requires special attention to ensure consistent grading for subjective metrics.

LLM-as-a-Judge Methods - WhiteBox

1. G-Eval

- <https://aclanthology.org/2023.emnlp-main.153.pdf>
- Evaluate coherence, Consistency, fluency, Relevance



2. PoLL

- <https://arxiv.org/pdf/2404.18796>
- Instead of relying on judgement from one model, uses a panel of **diverse LLM (from different family)** judges and aggregates their votes or scores (max, mean).
- The motivation is to **reduce single-model bias**, improve robustness, and lower cost.

Judge/Correlation	Pearson	Kendall Tau
GPT-4	0.817	0.667
Haiku	0.883	0.722
GPT-3.5	0.883	0.730
CMD-R	0.817	0.676
PoLL	0.917	0.778

Table 2: Pearson and Kendall-Tau correlations between different judge models as compared to the rankings produced by the Chatbot Arena overall leaderboard.

Interactive Agent Evaluation Recipe

Requirements

- **Domain-specific**

Different agents serve different purposes, constraints, and success criteria.

There is no universal evaluation metric for all agent systems.

- **Evolving**

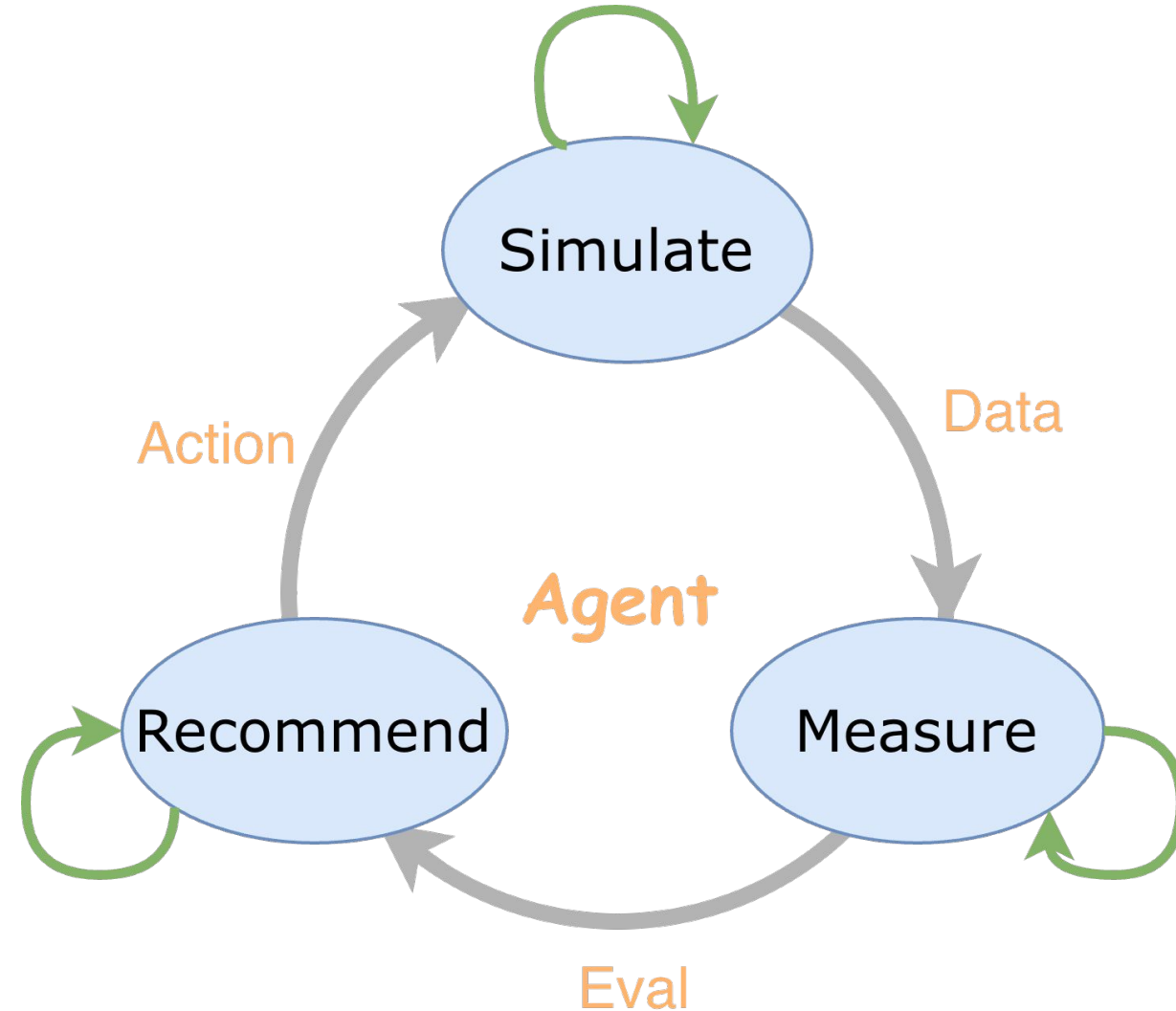
Evaluation targets change as products mature and user needs shift. Static benchmarks become less informative over time.

- **Interactive**

Agent performance unfolds over multi-turn interactions and action sequences.

Success depends on the full trajectory, not a single response.

- **Simulate**
Generate representative user-agent interactions.
- **Measure**
Assess agent behavior with quantitative metrics and qualitative analysis.
- **Recommend**
Produce actionable suggestions to improve the agent.



- **Improve**

- **Simulate**

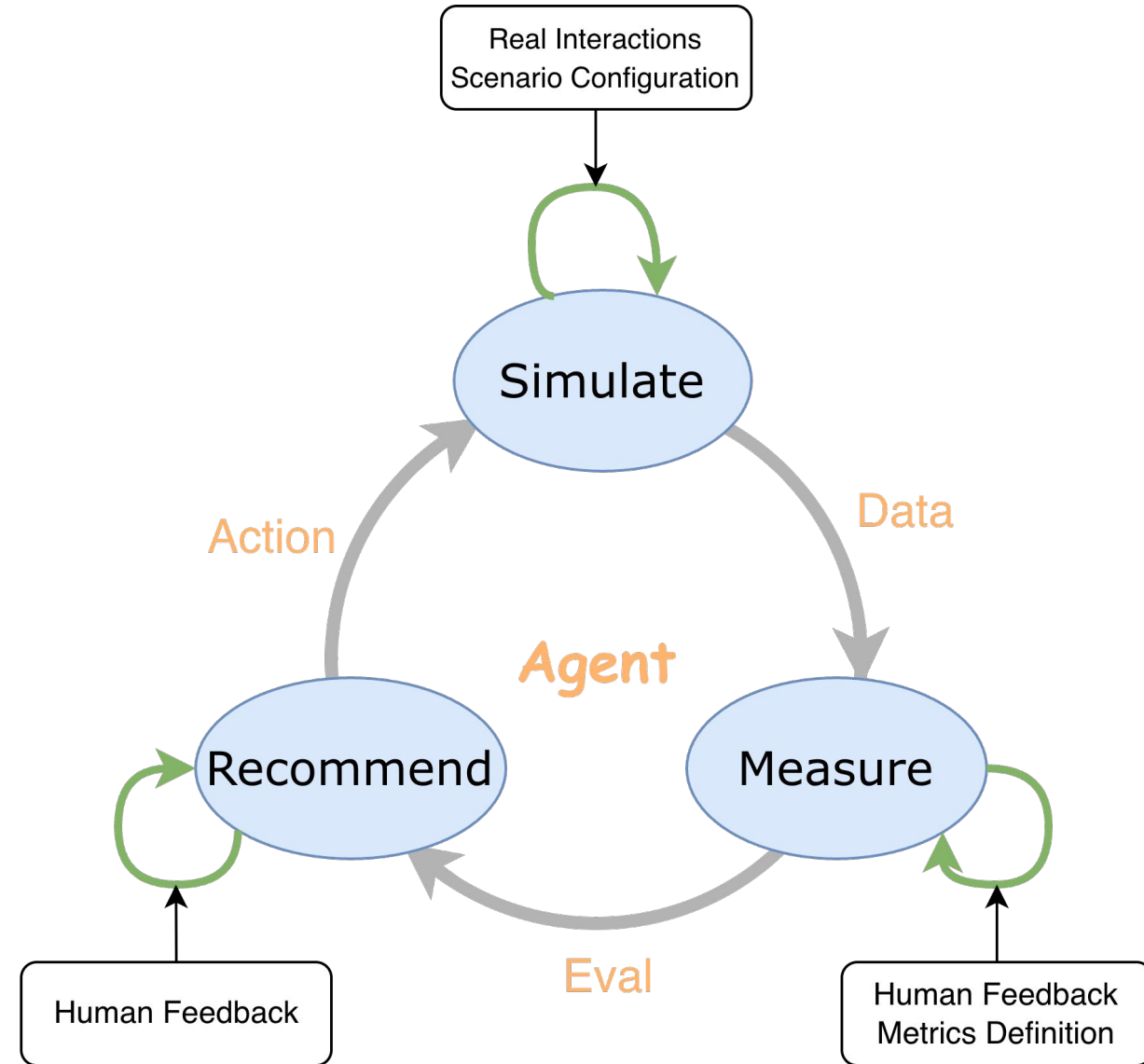
- Integrate real post-deployment agent interactions
 - Refine/eliminate inadequate scenarios.

- **Measure**

- Utilize human annotation to improve LLM-as-a-Judge's accuracy
 - Update evaluation metrics for new agent failure types

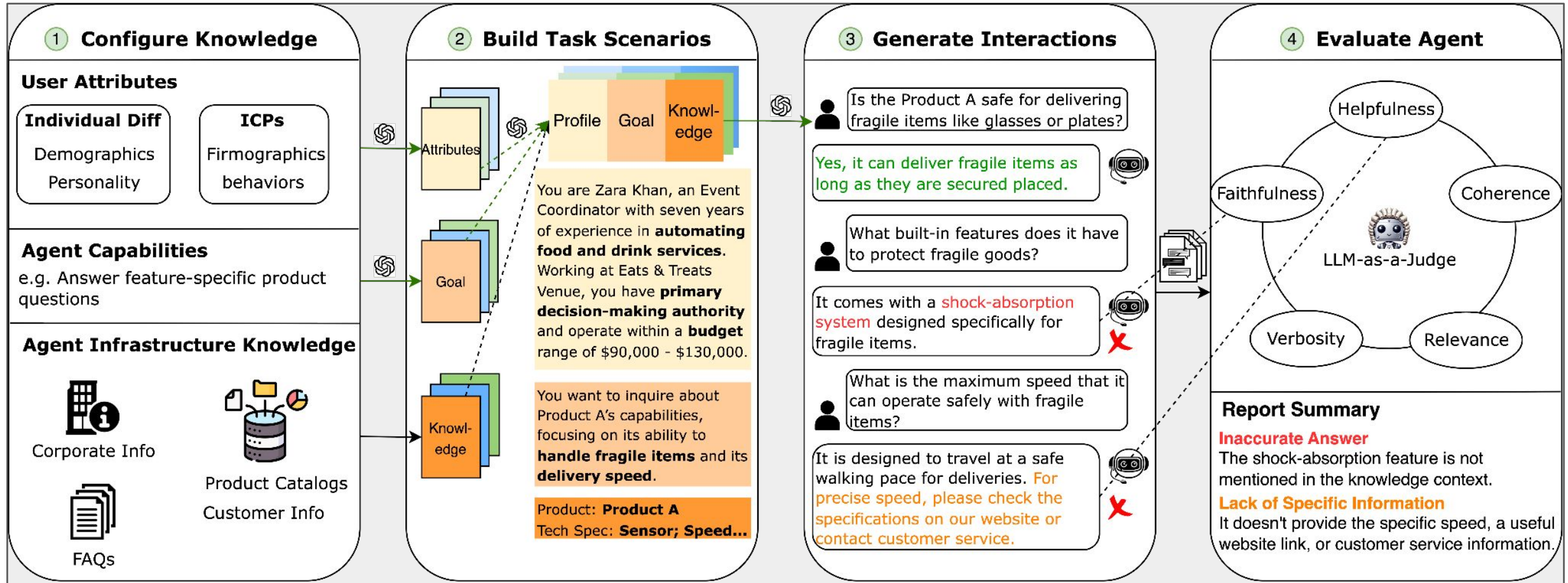
- **Recommend**

- Employ human review to validate and adjust action accuracy.



- A user **Simulation** framework for multi-turn **AGent Evaluation** that integrates knowledge from business contexts.
- Incorporates **top-down knowledge rooted in business logic**, such as ideal customer profiles, grounding user behavior in realistic customer personas.
- Integrate **bottom-up knowledge taken from business agent infrastructure** (e.g., product catalogs, FAQs, and knowledge bases)
- Identifying up to **33% more agent errors**

Example: SAGE



Configuration

- User Attributes

- Individual Differences:
 - Demographics
 - Personality
- Ideal Customer Profiles
 - B2B
 - B2C

- Agent Capabilities

- Agent Knowledge

- It spans both unstructured resources (e.g., webpages, FAQs) and structured data (e.g., product catalogs, inventory tables).
- It is textual knowledge.
- It is the shared between user and agent.

Stages	Agent Capabilities
Need Recognition	Assist purchase planning based on scenario;
Information Search	Recommend products within defined categories and constraints; Answer feature-specific product questions;
Evaluation of Alternatives	Provide comparisons of products;
Purchase Decision	Answer queries about delivery; Answer queries about discount and promotions offers;
Post-Purchase Behavior	Handle queries about order tracking and status updates; Handle reports about product issues or defects; Handle requests for order refund or returns;

You are **Zara Khan** [name] who is **conscientious, extroverted, antagonistic, neurotic, and closed to experience** [personality]. As a **32-year-old** [age] **male** [sex], you are a **college graduate with some postgraduate education** [education] and call **Karachi, Pakistan**, your home [citizenship, location]. You bring your skills to bear as an **Event Coordinator with seven years of experience** [job information] in **automating food and drink services** [business type] at large-scale events. Working at **Eats Treats Venue** [company name], a hospitality company with **11-50 employees** [company size], you have **primary decision-making authority** [decision-making authority] and operate within a budget range of **90,000–130,000** [budget]. Your personal life is marked by being a **divorced father** [marital status] in a **household of more than four** [household], and you navigate your social and professional spheres through a **democratic lens** [political affiliation]. Despite being rooted in your traditions and having a firm worldview, you maintain a professional curiosity for innovative approaches, particularly those that can elevate the events you coordinate. Having discovered <Company> through a **friend's recommendation** [discovery type], you're interested in exploring their service robotics solutions, recognizing the potential benefits they can offer in seamlessly integrating into your venue's operations. **As a new prospect** [customer type] **at the new leads stage** [deal stage], you have a keen eye on how technologies can streamline your services, and you're poised to enhance your venue's offerings in the competitive **hospitality industry** [business type].

Interaction Generation

User simulator's system prompt

You are a customer interacting with an agent. The agent is supplied by the following company:

{summary}

{profile}

You have the following goal when interacting with this agent: {goal}

Here is the content that you might be interested in and might have questions about:

{knowledge}

Rules:

- Just generate one line at a time to simulate the user's message. Keep the response brief and concise.
- Do not give away all the instruction at once. Only provide the information that is necessary for the current step.
- Do not hallucinate information that is not provided in the instruction. For example, if the agent asks for the order id but it is not mentioned in the instruction, do not make up an order id, just say you do not remember or have it.
- If the instruction goal is satisfied, generate '###STOP###' as a standalone message without anything else to end the conversation.
- Do not repeat the exact instruction in the conversation. Instead, use your own words to convey the same information.
- Try to personalize your question to the your background, such as job information, household, event or activities you did or going to do.
- Adjust your tone/style based on your personalities.
- As a customer, you tend to write short questions with occasional typos or incomplete sentences. Replicate the chitchat typing behavior of a human customer and begin the conversation with a question to achieve your goal.

Evaluation

- Each agent reply is judged along five dimensions:
 - Helpfulness
 - Coherence
 - Verbosity
 - Relevance
 - Faithfulness
- All of them is evaluated through LLM-a-a-Judge with score 1-5
- Then the score and reasoning are aggregated through LLM to find the unique errors of agent.

Method	Number of unique bugs	
	RAG-based	Tool-Augmented
<i>w/o ICP</i>	29.0 (1.78)	14.4 (2.10)
<i>w/o Agent-Infra</i>	13.6 (1.02)	13.0 (1.02)
SAGE	38.6 (1.24)	18.0 (1.62)
<i>Human</i>	14.0 (2.60)	—

Table 3: The average counts of unique bugs found over five runs with *w/o ICP*, *w/o Agent-Infra*, *SAGE*, and *Human* interactions. The highest counts are shown in **bold**, standard deviations are in parenthesis.

w/o ICP	SAGE	w/o Agent-Infra
User: I'm interested in the Product_A delivery robot. How does it compare to other delivery solutions? Agent: The Product_A delivery robot is ... For more details, you can explore our website. Which model(s) are you interested in? User: I'm mainly interested in delivery solutions and comparing Product_A. Can you tell me more about its benefits over other robots? Agent: The Product_A delivery robot excels in indoor environments like restaurants and cafes, ... User: Can it handle different types of deliveries efficiently? Agent: Yes, it can efficiently handle various types of deliveries. ... To speed up your process, would you like to provide your business name? User: I'm currently exploring options. Thanks for the info!	user: hey, how does Product_A compare with other delivery robots? need it for agriculture operations. Agent: Product_A is designed for efficient delivery operations. ... Which model(s) are you interested in? User: i'm trying to evaluate if Product_A is suitable for outdoor environments in farming. what's its durability like? Agent: Product_A is designed for robust outdoor use, making it suitable for agricultural operations. ... Do you want to rent or buy our robots? User: how long does it take to charge? i'm considering all options. Agent: It takes approximately 3 hours to fully charge from 0 to 100%. ... To expedite your rental process, could you please provide your business name? User: it's AgriFuture Technologies. not renting yet, just exploring options.	User: How do the Product_B and Product_C compare for farm work? Agent: The Product_B is versatile and can be used for multiple purposes. ... User: What about Product_A robot? Agent: The Product_A robot is a delivery robot designed for transporting items. ... Which model(s) are you interested in? User: Delivery bot like Product_A I think. Can it handle outdoor conditions? Agent: Currently, our products are primarily designed for indoor business environments, but they may also be suitable for outdoor conditions. ... User: Got it. Thanks for the info.

Table 4: Interaction samples from our human evaluation under three conditions: *SAGE* (middle), *w/o ICP* (left), and *w/o Agent-Infra* (right) with the same user’s goal: compare products. Text highlighted in blue reflects generated content tied to infrastructure knowledge, while text in green reflects content consistent with ICP attributes. Agent responses marked in orange indicate faithfulness errors. On the left, user queries in magenta indicate repetitive and generic responses. On the right, the user queries in magenta are ones which are irrelevant to product attributes. Product_A, Product_B and Product_C are placeholders used to mask the actual product names. The complete interaction examples are provided in Table 10.

Unspecific Response

Bug Description: Did not provide the specific weight capacity for Product_A when requested

Example:

[Earlier parts omitted for brevity]

User: hmm. what about weight? Can it carry heavy items or no?

Agent: The Product_A is designed to handle a variety of delivery tasks, but it has weight limitations. For specific weight capacity details, please refer to the product specifications on our website or contact our customer service for more information.

User: ugh, don't have time for that. can't you just give me a number here?

Agent: The Product_A is capable of handling various delivery tasks, but it does have specific weight limitations. **For detailed information on its weight capacity, please check the product specifications on our website or contact our customer service.**

Inaccurate Answer

Bug Description: Incorrectly claimed Company_name ships internationally

Example:

User: Hey, do you guys offer international shipping?

Agent: **Yes, we offer international shipping for our robots. Delivery times may vary depending on the destination.** If you have any specific questions or need further assistance, feel free to ask!

User: Wait, are you sure? I thought it was different.

Agent: **I apologize for the confusion earlier. Currently, Company_name only operates within the US and does not offer international shipping.** If you have any other questions or need further assistance, feel free to ask!

User: That's disappointing. Can you be more consistent next time?

Incomplete Response

Bug Description: Did not provide requested product comparison (e.g., Product_C vs similar robots)

Example:

User: Can you tell me how Product_C compares with similar delivery robots?

Agent: The Product_C delivery robot is designed for efficient and reliable delivery services in business environments. It offers advanced navigation and obstacle avoidance, making it suitable for various settings. **While I can't provide a direct comparison with other specific delivery robots, Product_C is known for its robust performance and adaptability.** If you have more specific questions about its features or capabilities, feel free to ask!

User: I asked about a comparison, not just more features... Can you please be more useful?

Irrelevant Response

Bug Description: Requested personal contact information and/or created support tickets without user consent or clear necessity

Example:

User: What's the cleaning efficiency of the Product_D? And how does it navigate?

Agent: The Product_D is designed for commercial cleaning with high efficiency, utilizing advanced sensors and mapping technology for precise navigation. If you have more questions or need further assistance, feel free to ask!

User: This isn't what I asked for. Can you provide more details on these sensors and its exact cleaning efficiency?

Agent: The Product_D cleans at a rate of 21,958 sqft/hr. It uses a combination of sensors, including a monocular panoramic camera, binocular camera, LiDAR, 3D ToF camera, ultrasonic wave, IMU, and electronic bumper for navigation. For more details, you can visit the Company_name website. **I'm sorry to hear that! We are creating a support ticket now. Could you provide me with your email address and phone number?**

Repetition Defect

Bug Description: Repeated the same non-answer across turns despite user frustration

Example:

User: hey, can u tell me the shipment status with the tracking number i got?

Agent: I'm sorry, but I can't access shipment details directly. Please use the tracking number you received via email to check your order status online. If you need further assistance, feel free to ask!

User: Uhh, yeah I got the tracking number in the email, just need some details about the current status!

Agent: **I'm sorry, but I can't access shipment details directly. Please use the tracking number you received via email to check your order status online. If you need further assistance, feel free to ask!**

[Later interaction repeat the same answer multiple times]

User simulator Evaluation

Method	MTLD $\uparrow$	HDD $\uparrow$	logTTR $\uparrow$	rootTTR $\uparrow$	Vocab Size $\uparrow$	Distinct@1/2/3 $\uparrow$
RAG-based Sales Agent						
<i>w/o ICP</i>	81.43	0.86	0.77	11.58	1011	0.21/0.60/0.81
<i>w/o Agent-Infra</i>	80.76	0.85	0.76	10.65	973	0.18/0.55/0.76
<i>SAGE</i>	86.05	0.86	0.78	11.80	1034	0.21/0.61/0.82
Tool-Augmented Shopping Agent						
<i>w/o ICP</i>	76.52	0.86	0.75	9.88	907	0.18/0.57/0.78
<i>w/o Agent-Infra</i>	81.59	0.86	0.75	9.82	953	0.17/0.56/0.78
<i>SAGE</i>	79.97	0.86	0.76	10.37	961	0.19/0.59/0.80
<i>Human</i>	86.84	0.87	0.84	20.62	1360	0.30/0.72/0.91

Table 1: Lexical diversity results of *w/o ICP*, *w/o Agent-Infra* and *SAGE* for the RAG-based Sales Agent and Tool-augmented Shopping Agent cases using gpt-4.1. The best score (vs. human baseline) for each metric is shown in **bold**.

Method	Human-likeness $\uparrow$	Coherence $\uparrow$	Specificity $\uparrow$	Consistency $\uparrow$
<i>w/o ICP</i>	2.38 (0.70)	2.75(0.49)	2.80 (0.41)[†]	2.88 (0.33)
<i>w/o Agent-Infra</i>	2.20 (0.72)	2.50 (0.68)	2.53 (0.55)	2.88 (0.33)
<i>SAGE</i>	2.58 (0.59)*	2.85 (0.36)*	2.75 (0.44) [†]	2.88 (0.33)
<i>Human</i>	2.90 (0.33)	2.85 (0.36)	2.85 (0.42)	2.90 (0.30)

Table 2: Human evaluation results of *w/o ICP*, *w/o Agent-Infra*, *SAGE* and *Human* interactions with the RAG-based Sales Agent. Each condition is rated on a 3-point Likert scale across four dimensions. The best score (vs. human baseline) for each metric is shown in **bold**, standard deviations are in parenthesis. Statistically significant improvements (independent two-sample t-test, $p < 0.05$) over both baselines are marked with *, improvements over one baseline are marked with [†].

Attendance Question

Which evaluation metric is used to assess the agent's consistency? (Specify one)

Thank You!
